From Science to Autoformalization

Science as the Search for Mathematical Laws

From Observation to Law

- **Johannes Kepler**: data $\rightarrow$ precise mathematical laws
- **Galileo**: nature is written in mathematics
- **Newton**: unified laws into predictive theory

Science becomes:

$$\text{Observation} \rightarrow \text{Pattern} \rightarrow \text{Mathematical Law}$$

Explanation = Mathematical Compression

- simple symbolic laws explain many phenomena
- emphasis on interpretability and structure

Phenomenology as a Scientific Style

- fast loop: experiment $\rightarrow$ regularity $\rightarrow$ effective law
- Kepler as early example
- today: particle phenomenology (collider data $\rightarrow$ EFT models)

From Mathematical Laws to Logic and Computation

Formal Foundations

- Frege, Russell, Hilbert: mathematics as symbolic logic
- proofs = explicit rule-based derivations

Computation = Mechanized Logic

- logical reasoning becomes executable
- programs $\leftrightarrow$ proofs

Formal Proof at Scale

- Kepler conjecture: computation-heavy proof
- Flyspeck: fully formal, machine-verified proof

Toward Formalized Physics

- growing formal libraries in mathematics
- long-term goal: formal treatments of physical theories
- e.g. Yang–Mills as a candidate milestone

$$\text{Mathematics} \rightarrow \text{Logic} \rightarrow \text{Computation} \rightarrow \text{Verification}$$

Autoformalization, Explainable AI, and Verified Reasoning

Autoformalization

- informal math/science $\rightarrow$ formal symbolic systems
- bridge human reasoning and machine verification

AI and the Return of Phenomenology

- LLMs: data $\rightarrow$ patterns $\rightarrow$ implicit "laws"
- but often non-symbolic and non-explainable

Explainable AI vs Black-Box Models

- science values interpretable, symbolic laws
- modern AI often produces opaque predictions

The Verification Problem

- LLMs generate programs, proofs, and mathematics
- outputs can be plausible but incorrect ("AI slop")
- need for formal verification and proof assistants

Emerging Synthesis

$$\text{Data} \rightarrow \text{AI Generation} \rightarrow \text{Formalization} \rightarrow \text{Verification}$$